```
entry:
%0 = icmp eq i32 %num_args, 3, !dbg !15
 br i1 %0, label %assert_end, label %assert_fail, !dbg !15, !prof !16
T | F

assert_end:
%.not = icmp eq ptr %args, null, !dbg !15
 br i1 %.not, label %assert_fail1, label %assert_end2, !dbg !15, !prof !20
T | F

assert_fail:
%1 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %1(ptr nonnull @.str.1, ptr nonnull @.str), !dbg !15
 br label %common.ret, !dbg !15

assert_end2:
%var_A.type_index = load i32, ptr %args, align 4, !dbg !15
 %var_A.type_index.fr = freeze i32 %var_A.type_index, !dbg !15
 %3 = icmp sgt i32 %var_A.type_index.fr, 63, !dbg !15
 br i1 %3, label %assert_end4, label %switch.early.test, !dbg !15
T | F

switch.early.test:
switch i32 %var_A.type_index.fr, label %assert_fail3 [
 i32 7, label %assert_end4
 i32 4, label %assert_end4
 i32 0, label %assert_end4
 ], !dbg !15
def | 7 | 4 | 0

assert_end4:
%5 = getelementptr inbounds i8, ptr %args, i64 16, !dbg !15
 %var_B.type_index = load i32, ptr %5, align 4, !dbg !15
 %var_B.type_index.fr = freeze i32 %var_B.type_index, !dbg !15
 %6 = icmp sgt i32 %var_B.type_index.fr, 63, !dbg !15
 br i1 %6, label %assert_end6, label %switch.early.test67, !dbg !15
T | F

assert_fail1:
%2 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %2(ptr nonnull @.str.1, ptr nonnull @.str.2), !dbg !15
 br label %common.ret, !dbg !15

switch.early.test67:
switch i32 %var_B.type_index.fr, label %assert_fail5 [
 i32 7, label %assert_end6
 i32 4, label %assert_end6
 i32 0, label %assert_end6
 ], !dbg !15
def | 7 | 4 | 0

assert_end6:
%8 = getelementptr inbounds i8, ptr %args, i64 32, !dbg !15
 %var_T_multiply.type_index = load i32, ptr %8, align 4, !dbg !15
 %var_T_multiply.type_index.fr = freeze i32 %var_T_multiply.type_index, !dbg
... !15
 %9 = icmp sgt i32 %var_T_multiply.type_index.fr, 63, !dbg !15
 br i1 %9, label %assert_end8, label %switch.early.test68, !dbg !15
T | F

assert_fail3:
%4 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %4(ptr nonnull @.str.1, ptr nonnull @.str.3), !dbg !15
 br label %common.ret, !dbg !15

switch.early.test68:
switch i32 %var_T_multiply.type_index.fr, label %assert_fail7 [
 i32 7, label %assert_end8
 i32 4, label %assert_end8
 i32 0, label %assert_end8
 ], !dbg !15
def | 7 | 4 | 0

assert_end8:
%11 = getelementptr inbounds i8, ptr %args, i64 8, !dbg !15
 %12 = load ptr, ptr %11, align 8, !dbg !15
 %13 = icmp eq i32 %var_A.type_index.fr, 72, !dbg !15
 %var_A.idx = select i1 %13, i64 16, i64 0, !dbg !15
 %var_A = getelementptr inbounds i8, ptr %12, i64 %var_A.idx, !dbg !15
 %14 = getelementptr inbounds i8, ptr %args, i64 24, !dbg !15
 %15 = load ptr, ptr %14, align 8, !dbg !15
 %16 = icmp eq i32 %var_B.type_index.fr, 72, !dbg !15
 %var_B.idx = select i1 %16, i64 16, i64 0, !dbg !15
 %var_B = getelementptr inbounds i8, ptr %15, i64 %var_B.idx, !dbg !15
 %17 = getelementptr inbounds i8, ptr %args, i64 40, !dbg !15
 %18 = load ptr, ptr %17, align 8, !dbg !15
 %19 = icmp eq i32 %var_T_multiply.type_index.fr, 72, !dbg !15
 %var_T_multiply.idx = select i1 %19, i64 16, i64 0, !dbg !15
 %var_T_multiply = getelementptr inbounds i8, ptr %18, i64
... %var_T_multiply.idx, !dbg !15
 %.not69 = icmp eq ptr %12, null, !dbg !15
 br i1 %.not69, label %assert_fail9, label %assert_end10, !dbg !15, !prof !20
T | F

assert_end10:
%21 = getelementptr inbounds i8, ptr %var_A, i64 16, !dbg !15
 %22 = load i32, ptr %21, align 4, !dbg !15
 %23 = icmp eq i32 %22, 2, !dbg !15
 br i1 %23, label %assert_end12, label %assert_fail11, !dbg !15, !prof !16
T | F

assert_end12:
%25 = getelementptr inbounds i8, ptr %var_A, i64 24, !dbg !15
 %mul.var_A.shape = load ptr, ptr %25, align 8, !dbg !15
 %26 = load i64, ptr %mul.var_A.shape, align 8, !dbg !15, !tbaa !30
 %M = trunc i64 %26 to i32, !dbg !15
 %27 = getelementptr inbounds i8, ptr %mul.var_A.shape, i64 8, !dbg !15
 %28 = load i64, ptr %27, align 8, !dbg !15, !tbaa !41
 %N = trunc i64 %28 to i32, !dbg !15
 %29 = getelementptr inbounds i8, ptr %var_A, i64 32, !dbg !15
 %mul.var_A.strides = load ptr, ptr %29, align 8, !dbg !15
 %30 = getelementptr inbounds i8, ptr %var_A, i64 12, !dbg !15
 %dev_id = load i32, ptr %30, align 4, !dbg !15
 %A = load ptr, ptr %var_A, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %A, i64 64) ], !dbg !15
 %.not70 = icmp eq ptr %15, null, !dbg !15
 br i1 %.not70, label %assert_fail13, label %assert_end14, !dbg !15, !prof !20
T | F

assert_fail11:
%24 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %24(ptr nonnull @.str.1, ptr nonnull @.str.7), !dbg !15
 br label %common.ret, !dbg !15

assert_fail13:
%31 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %31(ptr nonnull @.str.1, ptr nonnull @.str.8), !dbg !15
 br label %common.ret, !dbg !15

assert_end14:
%32 = getelementptr inbounds i8, ptr %var_B, i64 16, !dbg !15
 %33 = load i32, ptr %32, align 4, !dbg !15
 %34 = icmp eq i32 %33, 2, !dbg !15
 br i1 %34, label %assert_end16, label %assert_fail15, !dbg !15, !prof !16
T | F

assert_end16:
%36 = getelementptr inbounds i8, ptr %var_B, i64 24, !dbg !15
 %mul.var_B.shape = load ptr, ptr %36, align 8, !dbg !15
 %37 = getelementptr inbounds i8, ptr %var_B, i64 32, !dbg !15
 %mul.var_B.strides = load ptr, ptr %37, align 8, !dbg !15
 %B = load ptr, ptr %var_B, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %B, i64 64) ], !dbg !15
 %.not71 = icmp eq ptr %18, null, !dbg !15
 br i1 %.not71, label %assert_fail17, label %assert_end18, !dbg !15, !prof !20
T | F

assert_fail15:
%35 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %35(ptr nonnull @.str.1, ptr nonnull @.str.9), !dbg !15
 br label %common.ret, !dbg !15

assert_fail17:
%38 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %38(ptr nonnull @.str.1, ptr nonnull @.str.10), !dbg !15
 br label %common.ret, !dbg !15

assert_end18:
%39 = getelementptr inbounds i8, ptr %var_T_multiply, i64 16, !dbg !15
 %40 = load i32, ptr %39, align 4, !dbg !15
 %41 = icmp eq i32 %40, 2, !dbg !15
 br i1 %41, label %assert_end20, label %assert_fail19, !dbg !15, !prof !16
T | F

assert_end20:
%43 = getelementptr inbounds i8, ptr %var_T_multiply, i64 24, !dbg !15
 %mul.var_T_multiply.shape = load ptr, ptr %43, align 8, !dbg !15
 %44 = getelementptr inbounds i8, ptr %var_T_multiply, i64 32, !dbg !15
 %mul.var_T_multiply.strides = load ptr, ptr %44, align 8, !dbg !15
 %T_multiply = load ptr, ptr %var_T_multiply, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %T_multiply, i64 64) ], !dbg
... !15
 %45 = getelementptr inbounds i8, ptr %var_A, i64 22, !dbg !15
 %46 = load i16, ptr %45, align 2, !dbg !15
 %47 = icmp eq i16 %46, 1, !dbg !15
 %48 = getelementptr inbounds i8, ptr %var_A, i64 21, !dbg !15
 %49 = load i8, ptr %48, align 1, !dbg !15
 %50 = icmp eq i8 %49, 32, !dbg !15
 %51 = getelementptr inbounds i8, ptr %var_A, i64 20, !dbg !15
 %52 = load i8, ptr %51, align 1, !dbg !15
 %53 = icmp eq i8 %52, 2, !dbg !15
 %54 = and i1 %50, %53, !dbg !15
 %55 = and i1 %47, %54, !dbg !15
 br i1 %55, label %assert_end22, label %assert_fail21, !dbg !15, !prof !16
T | F

assert_fail19:
%42 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %42(ptr nonnull @.str.1, ptr nonnull @.str.11), !dbg !15
 br label %common.ret, !dbg !15

assert_end22:
%.not72 = icmp eq ptr %mul.var_A.strides, null, !dbg !15
 br i1 %.not72, label %if_end, label %if_then, !dbg !15, !prof !20
T | F

assert_fail21:
%56 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %56(ptr nonnull @.str.1, ptr nonnull @.str.12), !dbg !15
 br label %common.ret, !dbg !15

if_then:
%57 = load i64, ptr %mul.var_A.strides, align 8, !dbg !15, !tbaa !55
 %58 = trunc i64 %57 to i32, !dbg !15
 %59 = icmp eq i32 %N, %58, !dbg !15
 %60 = icmp eq i32 %M, 1, !dbg !15
 %61 = or i1 %60, %59, !dbg !15
 %62 = getelementptr inbounds i8, ptr %mul.var_A.strides, i64 8, !dbg !15
 %63 = load i64, ptr %62, align 8, !dbg !15, !tbaa !65
 %64 = and i64 %63, 4294967295, !dbg !15
 %65 = icmp eq i64 %64, 1, !dbg !15
 %66 = icmp eq i32 %N, 1, !dbg !15
 %67 = or i1 %66, %65, !dbg !15
 %68 = and i1 %61, %67, !dbg !15
 br i1 %68, label %if_end, label %assert_fail23, !dbg !15, !prof !16
T | F

assert_fail5:
%7 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %7(ptr nonnull @.str.1, ptr nonnull @.str.4), !dbg !15
 br label %common.ret, !dbg !15

if_end:
%69 = getelementptr inbounds i8, ptr %var_A, i64 40, !dbg !15
 %70 = load i64, ptr %69, align 8, !dbg !15
 %71 = icmp eq i64 %70, 0, !dbg !15
 br i1 %71, label %assert_end26, label %assert_fail25, !dbg !15, !prof !16
T | F

assert_fail23:
%72 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %72(ptr nonnull @.str.1, ptr nonnull @.str.13), !dbg !15
 br label %common.ret, !dbg !15

assert_end26:
%74 = getelementptr inbounds i8, ptr %var_A, i64 8, !dbg !15
 %75 = load i32, ptr %74, align 4, !dbg !15
 %76 = icmp eq i32 %75, 1, !dbg !15
 br i1 %76, label %assert_end28, label %assert_fail27, !dbg !15, !prof !16
T | F

assert_fail25:
%73 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %73(ptr nonnull @.str.1, ptr nonnull @.str.14), !dbg !15
 br label %common.ret, !dbg !15

assert_end28:
%78 = icmp ne ptr %A, null, !dbg !15
 %79 = mul nsw i32 %N, %M, !dbg !15
 %80 = icmp eq i32 %79, 0, !dbg !15
 %81 = or i1 %80, %78, !dbg !15
 br i1 %81, label %assert_end30, label %assert_fail29, !dbg !15, !prof !16
T | F

assert_fail27:
%77 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %77(ptr nonnull @.str.1, ptr nonnull @.str.15), !dbg !15
 br label %common.ret, !dbg !15

assert_end30:
%83 = getelementptr inbounds i8, ptr %var_B, i64 22, !dbg !15
 %84 = load i16, ptr %83, align 2, !dbg !15
 %85 = icmp eq i16 %84, 1, !dbg !15
 %86 = getelementptr inbounds i8, ptr %var_B, i64 21, !dbg !15
 %87 = load i8, ptr %86, align 1, !dbg !15
 %88 = icmp eq i8 %87, 32, !dbg !15
 %89 = getelementptr inbounds i8, ptr %var_B, i64 20, !dbg !15
 %90 = load i8, ptr %89, align 1, !dbg !15
 %91 = icmp eq i8 %90, 2, !dbg !15
 %92 = and i1 %88, %91, !dbg !15
 %93 = and i1 %85, %92, !dbg !15
 br i1 %93, label %assert_end32, label %assert_fail31, !dbg !15, !prof !16
T | F

assert_fail29:
%82 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %82(ptr nonnull @.str.1, ptr nonnull @.str.16), !dbg !15
 br label %common.ret, !dbg !15

assert_end32:
%95 = load i64, ptr %mul.var_B.shape, align 8, !dbg !15, !tbaa !67
 %96 = trunc i64 %95 to i32, !dbg !15
 %97 = icmp eq i32 %M, %96, !dbg !15
 br i1 %97, label %assert_end34, label %assert_fail33, !dbg !15, !prof !16
T | F

assert_fail31:
%94 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %94(ptr nonnull @.str.1, ptr nonnull @.str.17), !dbg !15
 br label %common.ret, !dbg !15

assert_end34:
%99 = getelementptr inbounds i8, ptr %mul.var_B.shape, i64 8, !dbg !15
 %100 = load i64, ptr %99, align 8, !dbg !15, !tbaa !77
 %101 = trunc i64 %100 to i32, !dbg !15
 %102 = icmp eq i32 %N, %101, !dbg !15
 br i1 %102, label %assert_end36, label %assert_fail35, !dbg !15, !prof !16
T | F

assert_fail33:
%98 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %98(ptr nonnull @.str.1, ptr nonnull @.str.18), !dbg !15
 br label %common.ret, !dbg !15

assert_end36:
%.not73 = icmp eq ptr %mul.var_B.strides, null, !dbg !15
 br i1 %.not73, label %if_end38, label %if_then37, !dbg !15, !prof !20
T | F

assert_fail35:
%103 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %103(ptr nonnull @.str.1, ptr nonnull @.str.19), !dbg !15
 br label %common.ret, !dbg !15

if_then37:
%104 = load i64, ptr %mul.var_B.strides, align 8, !dbg !15, !tbaa !79
 %105 = trunc i64 %104 to i32, !dbg !15
 %106 = icmp eq i32 %N, %105, !dbg !15
 %107 = icmp eq i32 %M, 1, !dbg !15
 %108 = or i1 %107, %106, !dbg !15
 %109 = getelementptr inbounds i8, ptr %mul.var_B.strides, i64 8, !dbg !15
 %110 = load i64, ptr %109, align 8, !dbg !15, !tbaa !89
 %111 = and i64 %110, 4294967295, !dbg !15
 %112 = icmp eq i64 %111, 1, !dbg !15
 %113 = icmp eq i32 %N, 1, !dbg !15
 %114 = or i1 %113, %112, !dbg !15
 %115 = and i1 %108, %114, !dbg !15
 br i1 %115, label %if_end38, label %assert_fail39, !dbg !15, !prof !16
T | F

assert_fail7:
%10 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %10(ptr nonnull @.str.1, ptr nonnull @.str.5), !dbg !15
 br label %common.ret, !dbg !15

if_end38:
%116 = getelementptr inbounds i8, ptr %var_B, i64 40, !dbg !15
 %117 = load i64, ptr %116, align 8, !dbg !15
 %118 = icmp eq i64 %117, 0, !dbg !15
 br i1 %118, label %assert_end42, label %assert_fail41, !dbg !15, !prof !16
T | F

assert_fail39:
%119 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %119(ptr nonnull @.str.1, ptr nonnull @.str.20), !dbg !15
 br label %common.ret, !dbg !15

assert_end42:
%121 = getelementptr inbounds i8, ptr %var_B, i64 8, !dbg !15
 %122 = load i32, ptr %121, align 4, !dbg !15
 %123 = icmp eq i32 %122, 1, !dbg !15
 br i1 %123, label %assert_end44, label %assert_fail43, !dbg !15, !prof !16
T | F

assert_fail41:
%120 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %120(ptr nonnull @.str.1, ptr nonnull @.str.21), !dbg !15
 br label %common.ret, !dbg !15

assert_end44:
%125 = getelementptr inbounds i8, ptr %var_B, i64 12, !dbg !15
 %126 = load i32, ptr %125, align 4, !dbg !15
 %127 = icmp eq i32 %dev_id, %126, !dbg !15
 br i1 %127, label %assert_end46, label %assert_fail45, !dbg !15, !prof !16
T | F

assert_fail43:
%124 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %124(ptr nonnull @.str.1, ptr nonnull @.str.22), !dbg !15
 br label %common.ret, !dbg !15

assert_end46:
%129 = icmp ne ptr %B, null, !dbg !15
 %130 = or i1 %80, %129, !dbg !15
 br i1 %130, label %assert_end48, label %assert_fail47, !dbg !15, !prof !16
T | F

assert_fail45:
%128 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %128(ptr nonnull @.str.1, ptr nonnull @.str.23), !dbg !15
 br label %common.ret, !dbg !15

assert_end48:
%132 = getelementptr inbounds i8, ptr %var_T_multiply, i64 22, !dbg !15
 %133 = load i16, ptr %132, align 2, !dbg !15
 %134 = icmp eq i16 %133, 1, !dbg !15
 %135 = getelementptr inbounds i8, ptr %var_T_multiply, i64 21, !dbg !15
 %136 = load i8, ptr %135, align 1, !dbg !15
 %137 = icmp eq i8 %136, 32, !dbg !15
 %138 = getelementptr inbounds i8, ptr %var_T_multiply, i64 20, !dbg !15
 %139 = load i8, ptr %138, align 1, !dbg !15
 %140 = icmp eq i8 %139, 2, !dbg !15
 %141 = and i1 %137, %140, !dbg !15
 %142 = and i1 %134, %141, !dbg !15
 br i1 %142, label %assert_end50, label %assert_fail49, !dbg !15, !prof !16
T | F

assert_fail47:
%131 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %131(ptr nonnull @.str.1, ptr nonnull @.str.24), !dbg !15
 br label %common.ret, !dbg !15

assert_end50:
%144 = load i64, ptr %mul.var_T_multiply.shape, align 8, !dbg !15, !tbaa
... !91
 %145 = trunc i64 %144 to i32, !dbg !15
 %146 = icmp eq i32 %M, %145, !dbg !15
 br i1 %146, label %assert_end52, label %assert_fail51, !dbg !15, !prof !16
T | F

assert_fail49:
%143 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %143(ptr nonnull @.str.1, ptr nonnull @.str.25), !dbg !15
 br label %common.ret, !dbg !15

assert_end52:
%148 = getelementptr inbounds i8, ptr %mul.var_T_multiply.shape, i64 8,
... !dbg !15
 %149 = load i64, ptr %148, align 8, !dbg !15, !tbaa !101
 %150 = trunc i64 %149 to i32, !dbg !15
 %151 = icmp eq i32 %N, %150, !dbg !15
 br i1 %151, label %assert_end54, label %assert_fail53, !dbg !15, !prof !16
T | F

assert_fail51:
%147 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %147(ptr nonnull @.str.1, ptr nonnull @.str.26), !dbg !15
 br label %common.ret, !dbg !15

assert_end54:
%.not74 = icmp eq ptr %mul.var_T_multiply.strides, null, !dbg !15
 br i1 %.not74, label %if_end56, label %if_then55, !dbg !15, !prof !20
T | F

assert_fail53:
%152 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %152(ptr nonnull @.str.1, ptr nonnull @.str.27), !dbg !15
 br label %common.ret, !dbg !15

if_then55:
%153 = load i64, ptr %mul.var_T_multiply.strides, align 8, !dbg !15, !tbaa
... !103
 %154 = trunc i64 %153 to i32, !dbg !15
 %155 = icmp eq i32 %N, %154, !dbg !15
 %156 = icmp eq i32 %M, 1, !dbg !15
 %157 = or i1 %156, %155, !dbg !15
 %158 = getelementptr inbounds i8, ptr %mul.var_T_multiply.strides, i64 8,
... !dbg !15
 %159 = load i64, ptr %158, align 8, !dbg !15, !tbaa !113
 %160 = and i64 %159, 4294967295, !dbg !15
 %161 = icmp eq i64 %160, 1, !dbg !15
 %162 = icmp eq i32 %N, 1, !dbg !15
 %163 = or i1 %162, %161, !dbg !15
 %164 = and i1 %157, %163, !dbg !15
 br i1 %164, label %if_end56, label %assert_fail57, !dbg !15, !prof !16
T | F

assert_fail9:
%20 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %20(ptr nonnull @.str.1, ptr nonnull @.str.6), !dbg !15
 br label %common.ret, !dbg !15

if_end56:
%165 = getelementptr inbounds i8, ptr %var_T_multiply, i64 40, !dbg !15
 %166 = load i64, ptr %165, align 8, !dbg !15
 %167 = icmp eq i64 %166, 0, !dbg !15
 br i1 %167, label %assert_end60, label %assert_fail59, !dbg !15, !prof !16
T | F

assert_fail57:
%168 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %168(ptr nonnull @.str.1, ptr nonnull @.str.28), !dbg !15
 br label %common.ret, !dbg !15

assert_end60:
%170 = getelementptr inbounds i8, ptr %var_T_multiply, i64 8, !dbg !15
 %171 = load i32, ptr %170, align 4, !dbg !15
 %172 = icmp eq i32 %171, 1, !dbg !15
 br i1 %172, label %assert_end62, label %assert_fail61, !dbg !15, !prof !16
T | F

assert_fail59:
%169 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %169(ptr nonnull @.str.1, ptr nonnull @.str.29), !dbg !15
 br label %common.ret, !dbg !15

assert_end62:
%174 = getelementptr inbounds i8, ptr %var_T_multiply, i64 12, !dbg !15
 %175 = load i32, ptr %174, align 4, !dbg !15
 %176 = icmp eq i32 %dev_id, %175, !dbg !15
 br i1 %176, label %assert_end64, label %assert_fail63, !dbg !15, !prof !16
T | F

assert_fail61:
%173 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %173(ptr nonnull @.str.1, ptr nonnull @.str.30), !dbg !15
 br label %common.ret, !dbg !15

assert_end64:
%178 = icmp ne ptr %T_multiply, null, !dbg !15
 %179 = or i1 %80, %178, !dbg !15
 br i1 %179, label %assert_end66, label %assert_fail65, !dbg !15, !prof !16
T | F

assert_fail63:
%177 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %177(ptr nonnull @.str.1, ptr nonnull @.str.31), !dbg !15
 br label %common.ret, !dbg !15

assert_end66:
tail call fastcc void @mul_compute_(i32 %M, i32 %N, ptr %T_multiply, ptr
... %A, ptr %B), !dbg !15
 br label %common.ret, !dbg !15

assert_fail65:
%180 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %180(ptr nonnull @.str.1, ptr nonnull @.str.32), !dbg !15
 br label %common.ret, !dbg !15

common.ret:
%common.ret.op = phi i32 [ -1, %assert_fail ], [ -1, %assert_fail1 ], [
... -1, %assert_fail3 ], [ -1, %assert_fail5 ], [ -1, %assert_fail7 ], [ -1,
... %assert_fail9 ], [ -1, %assert_fail11 ], [ -1, %assert_fail13 ], [ -1,
... %assert_fail15 ], [ -1, %assert_fail17 ], [ -1, %assert_fail19 ], [ -1,
... %assert_fail21 ], [ -1, %assert_fail23 ], [ -1, %assert_fail25 ], [ -1,
... %assert_fail27 ], [ -1, %assert_fail29 ], [ -1, %assert_fail31 ], [ -1,
... %assert_fail33 ], [ -1, %assert_fail35 ], [ -1, %assert_fail39 ], [ -1,
... %assert_fail41 ], [ -1, %assert_fail43 ], [ -1, %assert_fail45 ], [ -1,
... %assert_fail47 ], [ -1, %assert_fail49 ], [ -1, %assert_fail51 ], [ -1,
... %assert_fail53 ], [ -1, %assert_fail57 ], [ -1, %assert_fail59 ], [ -1,
... %assert_fail61 ], [ -1, %assert_fail63 ], [ -1, %assert_fail65 ], [ 0,
... %assert_end66 ]
 ret i32 %common.ret.op, !dbg !15
```

CFG for 'mul' function